

Harsh Raj

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Professional Summary

Final-year Computer Science student with experience developing AI-powered applications, developer tools, and full-stack web platforms. Skilled in Python, JavaScript/TypeScript, React, FastAPI, PostgreSQL, and cloud deployment, with research experience in automated code refactoring and software quality evaluation.

Education

VIT-AP University, Amaravati

B.Tech in Computer Science and Engineering, SCOPE

CGPA: 8.56 / 10

Expected: September 2026

Vidya Bharati Chinmaya Vidyalaya, Jamshedpur

Senior Secondary School

Class 10: 94.4% Class 12: 83.2%

March 2022

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS.

Frameworks & Libraries: React, Next.js, FastAPI, Flask, TailwindCSS, shadcn/ui.

Databases: PostgreSQL, Supabase (Postgres + RLS + Realtime), Drizzle ORM.

Tools & Platforms: Git, GitHub, GitLab CI/CD, Vitest, AWS, Vercel, Railway, Google Cloud, Sentry, PostHog, N8N.

AI/ML: Machine Learning, NLP, Google Gemini API, OpenAI API, Prompt Engineering.

Projects

• Argos - Competitive Live Debate Platform [Live Link]

Tech: Next.js 16, TypeScript, Supabase, PostgreSQL, Drizzle ORM, Gemini AI, TailwindCSS v4, shadcn/ui, Vercel

Built a full-stack competitive debate platform where two players argue timed rounds while an AI judge scores each argument across 4 dimensions and flags any of 10 logical fallacies in real time. Architected 25 API route handlers and 25+ Supabase migrations (Postgres with RLS) via Drizzle ORM, powering Elo-based ranked matchmaking, live spectating, audience voting, and Supabase Realtime score/turn sync. Integrated Google Gemini for automated scoring and feedback plus shareable OG result cards; instrumented with Sentry and PostHog, tested with Vitest in a GitLab CI pipeline, and deployed on Vercel with cron-driven daily topic generation and turn-timeout automation.

• MergeIQ - AI-Powered Pull Request Analyzer [GitHub] | [Live Link]

Tech: Python, FastAPI, Chrome Extension (MV3), Gemini 2.5 Flash, GitHub REST API, PyJWT, Railway

Built a Chrome extension and GitHub App that automatically analyzes every pull request using Gemini AI, delivering structured verdicts (Safe / Review Needed / Risky) with risk flags, summaries, and suggested review questions. Fetches live repo context (README, folder structure, dependencies, recent commits) for project-aware analysis. GitHub App mode posts analysis comments autonomously on PR open via webhook, requiring zero manual interaction from contributors.

• HealthAssist - AI Symptom Checker and Health Companion [Live Link]

Tech: Python, FastAPI, React, TailwindCSS, Google API, Generative AI, Render, PostgreSQL

Implemented a web-application that generates detailed analyses of the symptoms provided by the patient

including blood-report context-awareness, with personalized advices and precautions. Integrated 4 key data-points with 5 levels of analyses/advices.

- **CodeSage - AI code-base refactoring system and questions generator** [\[Live Link\]](#)

Tech: Python, Flask, React, TailwindCSS, Google API, Google Cloud, OpenAI API, LLMs

Implemented an AI code-base analysis and refactoring system that helps improve code quality, maintainability, and generates insights about your projects using Google's Gemini AI or other LLMs. Supports more than 15 programming languages. Conducted a detailed research report on the performance of 5 leading LLMs with different prompts and scoring metrics. Currently developing a cAST-based pipeline for cutting-edge refactoring capabilities.

Research

Performance Evaluation of LLMs and Prompt Strategies for Automated Python Code Refactoring:

Evaluated 4 frontier LLMs and 2 prompt-engineering strategies for automated Python code refactoring across 5 benchmark programs. Measured code quality and documentation improvements using BLEU, ROUGE-L, and Pylint metrics, achieving style-conformance scores of 9.2–9.7/10. Conducted comparative analysis of model and prompt performance to identify effective refactoring strategies.

Certifications

- **AWS Certified Cloud Practitioner (AWS)** [\[Link\]](#)
- **SAP Generative AI Developer (SAP)** [\[Link\]](#)
- **Data Analysis in R (DataCamp)** [\[Link\]](#)
- **MERN Full Stack (Ethnus)** [\[Link\]](#)
- **Oracle OCI Generative AI Professional (Oracle)** [\[Link\]](#)
- **Oracle Certified Foundations Associate (Oracle)** [\[Link\]](#)
- **JavaScript Essentials (Cisco)** [\[Link\]](#)
- **Networking Essentials (Cisco)** [\[Link\]](#)
- **Claude 101 (Anthropic)** [\[Link\]](#)
- **AI Fluency: Framework & Foundations (Anthropic)** [\[Link\]](#)

Achievements

- **2nd Place, The Return Journey Hackathon** (GDSC, IIT-Kanpur)

Leadership & Roles

- **Manager, TEAM NEXT NEXUS** (Prompt Engineering Club) July 2024 – December 2024
Led prompt-engineering study groups and reviewed generative-AI mini-project submissions, facilitating collaborative learning and technical discussions among club members.
- **Associate Manager, Madhya Bharat Association** (Cultural Club) July 2024–December 2024
Oversaw event logistics and team coordination for cultural events with 50+ participants.

Additional Information

Languages: English (Proficient), Hindi (Native)

Interests: AI Research, Number Theory, Art/Animation, Content Creation